Algebra Review: Unit 5 — Systems of Equations

Answer Key

Station 1: Checking Solutions and Types of Systems

1. Yes
2. No
3. Yes
4. $(2,\ 4)$
5. $(2,\ 1)$
6. $(4,\ 3)$
7. No solution *same slope, different y-intercepts (parallel lines)*
8. Infinitely many solutions *the second equation is 3 times the first (same line)*
9. Exactly one: $(2, 4)$ *slopes differ*
10. $k = 3$ *parallel lines need equal slopes*

Station 2: Solving Systems by Substitution

1. $(4,\ 8)$
2. $(4,\ 1)$
3. $(4,\ 1)$
4. $(2,\ 1)$
5. $(2,\ 3)$
6. $(3,\ 5)$
7. $(2,\ 8)$
8. $(3,\ 2)$
9. $(2,\ 3)$
10. 9 and 16 $x + y = 25,\ x = y + 7$

Station 3: Solving Systems by Elimination

1. $(6,\ 3)$
2. $(3,\ 5)$
3. $(4,\ 2)$
4. $(2,\ 3)$
5. $(3,\ 2)$
6. $(-1,\ 2)$
7. $(3,\ 1)$
8. $(1,\ 2)$
9. No solution *multiplying the first equation by 2 gives* $4x + 6y = 12 \neq 18$
10. 15 adult and 25 student tickets $a+s = 40,\ 8a+5s = 245$

Station 4: Choosing a Method and Systems in Context

1. $10 + 4v = 30 + 2v;\ v = 10$ visits, cost \$50
2. 15 nickels and 15 dimes $n + d = 30,\ 5n + 10d = 225$
3. boat 10 mph, current 2 mph $b + c = 12,\ b - c = 8$
4. 11 and 7
5. width 6 cm, length 9 cm
6. 7 and 28
7. notebook \$2, pen \$3 $5n + 3p = 19,\ 2n + 4p = 16$
8. $(2,\ 1)$
9. Infinitely many *substitution gives* $2 = 2$
10. No solution *substitution gives* $10 = 7$*, which is false*